

SELECTED SOLUTIONS TO PROBLEMS IN PDE

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1. HARMONIC FUNCTIONS

Problem 1.1. Let $\Omega \subseteq \mathbb{C}$ be open and let $u : \Omega \rightarrow \mathbb{C}$ be holomorphic in Ω . Prove that $\Re u$ and $\Im u$ are harmonic after identifying $\mathbb{C}$ with $\mathbb{R}^2$.

Proof. Since all partial derivatives of u exist, this is immediate from the Cauchy-Riemann equations. $\square$

Problem 1.2. Let $n \geq 2$ and denote by $\mathfrak{H}$ the real vector space of all harmonic functions $\Omega \rightarrow \mathbb{R}$. Prove that $\mathfrak{H}(\Omega)$ is infinite dimensional.

Proof. It is enough to prove the claim for $n = 2$ since $v(x) = u(x_1, x_2)$ is harmonic in $\mathbb{R}^n$ whenever $u(x_1, x_2)$ is harmonic in $\mathbb{R}^2$. With this in mind, we identify $\mathbb{R}^2$ with $\mathbb{C}$ and consider the embedding

$$\Phi : \ell^\infty(\mathbb{C}) \hookrightarrow \mathfrak{H}(\Omega), \quad (a_k)_{k=1}^\infty \xrightarrow{u} (z) := \Re \sum_{k=1}^\infty a_k n^{-k} z^n.$$

Note that the above is well defined since the real and imaginary parts of holomorphisms are harmonic. Since $\ell^\infty(\mathbb{C})$ is infinite dimensional over $\mathbb{R}$, the claim follows. $\square$

Problem 1.3. Let $\phi : (0, 1] \rightarrow \mathbb{R}$ be continuous and non-negative; assume that

$$\int_\delta^1 \phi(r) r^{n-1} dr \leq M \quad \forall \delta \in (0, 1),$$

for a constant $M > 0$ independent of δ . Define $u : B_1(0) \rightarrow \mathbb{R}$ by the rule $u(x) := \phi(|x|)$. Prove that $f \in L^1(B_1(0))$.

Proof. This is a straightforward computation using the change of variables theorem. First, by the monotone convergence theorem, we see that

$$\int_0^1 \phi(r) r^{n-1} dr = \lim_{\delta \searrow 0} \int_\delta^1 \phi(r) r^{n-1} dr < \infty.$$

Finally, observe that

$$\int_{B_1(0)} u(x) dm = \int_0^1 \int_{\partial B_r(0)} \phi(r) dS dr = n\omega_n \int_0^1 \phi(r) r^{n-1} dr < \infty.$$

$\square$

Problem 1.4. Let $\Omega \subseteq \mathbb{R}^n$ be a domain and let $u \in C^2(\Omega)$ be harmonic in Ω . If u achieves a local extremum, then u is constant.

Proof. Suppose that u achieves a local extremum m at a point $x_0 \in \Omega$. By replacing u with $-u$, we may assume that m is a local maximum. After a translation, we may also assume that $m = 0$. Next, we take $\varepsilon > 0$ so small that $B(x_0, \varepsilon) \Subset \Omega$ and $u(x) \leq m = 0$ on all of $B(x_0, \varepsilon)$. Since $u(x_0) = 0$ is a *global maximum* on the compact domain $\overline{B(x_0, \varepsilon)}$, the strong maximum principle forces u to be identically zero on all of Ω . Recalling that $u \in C^\omega(\Omega)$, the identity theorem implies that u is identically zero on Ω . $\square$

Problem 1.5. Deduce the strong maximum principle from Harnack's inequality.

Proof. Let $\Omega \subset \mathbb{R}^n$ be a non-empty bounded domain and let $u \in C^2(\Omega) \cap C(\bar{\Omega})$ be harmonic in the interior of Ω . Suppose that u achieves its global maximum $M := \sup_{\bar{\Omega}} u$ at a point $x_0 \in \Omega$. Let $V_j \nearrow \Omega$ be a sequence of connected open sets, each compactly contained in Ω and containing x_0 . For each fixed $j \in \mathbb{N}$, we may find by Harnack's inequality a constant $C > 0$ such that

$$\sup_{V_j} (M - u) \leq C \inf_{V_j} (M - u) = 0.$$

Thus, we have $0 \leq M - u \leq 0$ and so u is constant in V_j . Letting $j \nearrow \infty$ shows that u is constant in $\bar{\Omega}$. $\square$

Problem 1.6. Let $\Omega \subseteq \mathbb{R}^n$ be open and $u \in C(\Omega)$. Prove that u is subharmonic in Ω if and only if for every open ball $B \Subset \Omega$, and every harmonic function $v \in C^\infty(\Omega) \cap C(\bar{\Omega})$:

$$u \leq v \text{ on } \partial B \implies u \leq v \text{ in } B.$$

Proof. The forward implication is a consequence of the maximum principle for subharmonic functions. Conversely, we fix an open ball $B(x, r) \Subset \Omega$. Let v be the harmonic function on $B = B(x, r)$ obtained by convolving u with the Poisson kernel on ∂B . Since $u \leq v$ on $\partial \Omega$, we must have $u \leq v$ in all of B . In particular,

$$u(x) \leq v(x) = \int_{\partial B} v = \int_{\partial B} u.$$

This completes the proof. $\square$

Problem 1.7. Let $\Omega \subseteq \mathbb{R}^n$ be open and let u be harmonic in Ω . Show that $|\nabla u|^2$ is subharmonic in Ω .

Proof. We proceed by direct calculation. Put

$$v := |\nabla u|^2 = \sum_{j=1}^n u_{x_j}^2$$

in Ω and calculate for any $i = 1, \dots, n$:

$$v_{x_i} = \sum_{j=1}^n \left(2u_{x_j} u_{x_j x_i} \right).$$

Similarly,

$$v_{x_i x_i} = \sum_{j=1}^n \left(2u_{x_j x_i}^2 + 2u_{x_j} u_{x_j x_i x_i} \right).$$

Summing over $i = 1, \dots, n$, we discover that

$$\begin{aligned} \Delta v &= \sum_{i,j=1}^n \left(2u_{x_j x_i}^2 + 2u_{x_j} u_{x_j x_i x_i} \right) \geq 2 \sum_{i,j=1}^n u_{x_j} u_{x_j x_i x_i} \\ &= 2 \sum_{j=1}^n u_{x_j} (\Delta u)_{x_j} = 0. \end{aligned}$$

This completes the proof. $\square$

Problem 1.8. Let $u : \mathbb{R}^n \rightarrow \mathbb{R}$ be entire harmonic. If $f \in L^p(\mathbb{R}^n)$ for any $1 \leq p \leq \infty$, then f is constant. If $1 \leq p < \infty$, then $u \equiv 0$.

Proof. If $p = \infty$ this is Liouville's theorem. Henceforth, we assume that $1 \leq p < \infty$. Fix a point $x \in \mathbb{R}^n$ and let $R > 0$. We clearly have

$$|u(x)| \leq \frac{1}{\omega_n R^n} \int_{B_R(0)} |u| \leq \frac{C R^{n(1-\frac{1}{p})}}{R^n} \left(\int_{B_R(0)} |u|^p \right)^{1/p}.$$

Letting $R \rightarrow \infty$ yields the desired result. $\square$

Problem 1.9. Let $u : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a positive entire harmonic function. Prove that u is constant.

Proof. Fix two points $x, y \in \mathbb{R}^n$ and let $r \gg 1$ be given. Define $R := r + |x - y|$ so that

$$B(x, r) \subseteq B(y, R).$$

Then, we compute

$$u(x) = \frac{1}{\omega_n r^n} \int_{B(x,r)} u \leq \frac{1}{\omega_n r^n} \int_{B(x,R)} u = \frac{(r + |x - y|)^n}{r^n} u(y).$$

Letting $r \rightarrow \infty$ shows that $u(x) \leq u(y)$. Reversing the roles of x and y in the above shows that $u(x) = u(y)$. $\square$

Problem 1.10. Let $u : \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}$ be a positive harmonic function. Show that u is constant.

Proof. Without harm we may identify $\mathbb{R}^2$ with $\mathbb{C}$, and hence u is harmonic as a function $\mathbb{C}^\times \rightarrow \mathbb{R}$. Since $z \mapsto e^z$ is holomorphic in all of $\mathbb{C}$, we see that the composite

$$v(z) := u(e^z)$$

is a positive harmonic function $\mathbb{C} \rightarrow \mathbb{R}$. By the previous problem, v (and hence u) must be constant. $\square$

Problem 1.11. Let $\Omega \subseteq \mathbb{R}^n$ be a bounded open set and let $u : \Omega \rightarrow \mathbb{R}$ be bounded and harmonic. Prove that there exists a constant $C > 0$ such that

$$|\partial^\alpha u(x)| \leq \frac{Cn^{|\alpha|}}{\text{dist}(x, \partial\Omega)^{|\alpha|}}$$

for all multindices α and all $x \in \Omega$.

Proof. We argue by induction on the degree of α . Since u is bounded, the base case $|\alpha| = 0$ is trivial with $C := \sup_\Omega |u|$. Assume that the claim holds for all multindices α of order k . If β has order $k + 1$ there is $i = 1, \dots, n$ such that

$$\begin{aligned} \partial^\beta u(x) &= \partial^\alpha u_{x_i}(x) = \partial^\alpha \left(\frac{1}{\omega_n r^n} \int_{B(x,r)} u_{x_i} \right) \\ &= \partial^\alpha \left(\frac{1}{\omega_n r^n} \int_{\partial B(x,r)} u v^i \right) \\ &= \partial^\alpha \left(\frac{1}{\omega_n r} \int_{\partial B(0,1)} u(x + rz) v^i dS(z) \right) \\ &= \frac{1}{\omega_n r^n} \int_{\partial B(x,r)} \partial^\alpha u v^i \end{aligned}$$

where $B(x, r)$ is any ball compactly contained in Ω . Thus,

$$\left| \partial^\beta u(x) \right| \leq \frac{n\omega_n r^{n-1}}{\omega_n r^n} \cdot \frac{Cn^{|\alpha|}}{\text{dist}(x, \partial\Omega)^{|\alpha|}} = \frac{Cn^{|\beta|}}{r \text{dist}(x, \partial\Omega)^{|\alpha|}}.$$

Letting $r \nearrow \text{dist}(x, \partial\Omega)$ completes the inductive step. $\square$

Problem 1.12. Let $\mathbb{R}_+^n$ denote the half space $\{x_n > 0\}$ and assume that $u \in C^2(\mathbb{R}_+^n) \cap C(\overline{\mathbb{R}_+^n})$ is a bounded harmonic function in the open half space such that $u \leq 0$ on $\partial\mathbb{R}_+^n$. Prove that $u \leq 0$ in all of $\mathbb{R}_+^n$.

Proof. Let v, w be bounded harmonic functions in $\mathbb{R}_+^n$, both continuous up to the boundary. If $v \equiv w$ on $\partial\mathbb{R}_+^n$, then $v - w$ admits a bounded harmonic extension to $\mathbb{R}^n$. By Liouville's theorem, $v - w$ would be constant (i.e. identically zero) on $\mathbb{R}^n$. By this reasoning, we see that

$$u(x) := \frac{2x_n}{n\omega_n} \int_{\partial\mathbb{R}_+^n} \frac{u(y)}{|x - y|^n} dy$$

for all $x \in \mathbb{R}_+^n$. From this, it is obvious that $u \leq 0$ in $\mathbb{R}_+^n$. $\square$

Problem 1.13. Let $\mathbb{D}$ denote the open unit disk in $\mathbb{R}^2$. Construct a harmonic function $u : \mathbb{D} \rightarrow \mathbb{R}$ that is *not* uniformly continuous on $\mathbb{D}$.

Proof. Define a function $f : \partial\mathbb{D} = \mathbb{S}^1 \rightarrow \mathbb{R}$ by the following rule:

$$f(\cos t, \sin t) := \begin{cases} 1 & \text{if } 0 \leq t < \pi, \\ -1 & \text{if } \pi \leq t < 2\pi. \end{cases}$$

Then, f is continuous at all points save $(-1, 0)$ and $(1, 0)$. Let now $u : \mathbb{D} \rightarrow \mathbb{R}$ be the harmonic function obtained by convolving f with the Poisson kernel on $\partial\mathbb{D}$. If u were uniformly continuous on $\mathbb{D}$, we could extend it to a continuous function $v : \bar{\mathbb{D}} \rightarrow \mathbb{R}$. However, since f is continuous at all points except for $(-1, 0)$ and $(1, 0)$, we would have

$$f(\cos t, \sin t) := \begin{cases} 1 & \text{if } 0 < t < \pi, \\ -1 & \text{if } \pi < t < 2\pi \end{cases}$$

which is clearly impossible. $\square$

Problem 1.14. Let u be entire harmonic in $\mathbb{R}^n$ for $n \geq 2$. Establish the following:

- (1) At any point in $\mathbb{R}^n$, the tangent plane to the graph of u intersects the graph at more than one point.
- (2) Assume there exists a constant $C > 0$ and $m \in \mathbb{N}$

$$u(x) \geq -C(1 + |x|)^m$$

for all $x \in \mathbb{R}^n$. Then u is a polynomial of degree no larger than m .

Proof. We begin with (1). Fix a point $x_0 \in \mathbb{R}^n$ and let Γ describe the tangent plane to the graph of u at x_0 , i.e.

$$\Gamma(x) := u(x_0) + \nabla u(x_0) \cdot (x - x_0).$$

Clearly, Γ is entire harmonic in $\mathbb{R}^n$ and vanishes at x_0 . Suppose, by way of contradiction, that Γ only vanishes at x_0 . Let $B = B_\delta(x_0)$ be any large open ball. Since $B - \{x_0\}$ is path connected, $u - \Gamma$ must be either non-negative or non-positive on the whole of B . By slightly shrinking δ , we may assume that $u - \Gamma \geq 0$ on $\bar{B}$. But then, $u - \Gamma$ achieves its minimum in B and must be identically zero on $\bar{B}$. This contradicts the assumption that $u = \Gamma$ only at x_0 .

For the second part, let α be any multiindex with $|\alpha| > m$. Fix a point $x \in \mathbb{R}^n$ and let u be entire harmonic on $\mathbb{R}^n$. After a translation, we may assume that $u(x) = 0$. Then, for all $R > 0$:

$$0 = u(x) = \oint_{B(x, R)} u$$

whence we see that

$$\int_{B(x, R)} u^+ = \int_{B(x, R)} u^-.$$

Now, for a suitable (corrected) constant $C > 0$ independent of R we have

$$\begin{aligned} |\partial^\alpha u(x)| &= \frac{C}{R^{n+|\alpha|}} \int_{B(x, R)} |u| \, dm = \frac{2C}{R^{n+|\alpha|}} \int_{B(x, R)} u^- \leq \frac{C}{R^{n+|\alpha|}} \int_{B(x, R)} (1 + |y|)^m \, dy \\ &\leq \frac{C}{R^{n+|\alpha|}} \int_{B(x, R)} (1 + |x - y|)^m \, dy. \end{aligned}$$

Hence, it follows that

$$|\partial^\alpha u(x)| \leq \frac{C(1 + |R|)^m}{R^{n+|\alpha|}} R^n \xrightarrow{R \rightarrow \infty} 0.$$

This proves the claim. $\square$

Problem 1.15. Let $\Omega \subseteq \mathbb{R}^n$ be open and let $u \in C^1(\Omega)$. Suppose that for every $x \in \Omega$ there is $r^* > 0$ so that

$$\int_{\partial B(x,r)} \partial_\nu u \, dS = 0, \quad \forall r \in (0, r^*).$$

Show that $u \in C^\omega(\mathbb{R}^n)$ and that u is harmonic in Ω .

Proof. We show that u satisfies the local mean value property in Ω . Let $x \in \Omega$ and $r^* > 0$ as above. Define

$$\eta : (0, r^*) \rightarrow \mathbb{R}, \quad \eta(r) := \oint_{\partial B(x,r)} u \, dS.$$

Note that η may be extended to all of $[0, r^*)$ by putting $\eta(0) = u(x)$. For all $r \in (0, r^*)$, we find that

$$\begin{aligned} \eta'(r) &= \frac{d}{dr} \left(\frac{1}{n\omega_n r^{n-1}} \int_{\partial B(x,r)} u \, dS \right) \\ &= \frac{d}{dr} \left(\frac{1}{n\omega_n} \int_{\partial B(0,1)} u(x + rz) \, dS(z) \right) \\ &= \frac{1}{n\omega_n} \int_{\partial B(0,1)} \nabla u(x + rz) \cdot z \, dS(z) \\ &= \frac{1}{n\omega_n r^{n-1}} \int_{\partial B(x,r)} \nabla u(y) \cdot \nu(y) \, dS \\ &= \oint_{\partial B(x,r)} \partial_\nu u \, dS = 0. \end{aligned}$$

This shows that η is constant on $[0, r^*)$ and the proof is complete. $\square$

Problem 1.16. content...

2. FAMILIES OF HARMONIC FUNCTIONS & PERRON'S METHOD

Problem 2.1. Let Ω be a bounded domain and let $\{u_j\}_{j \in \mathbb{N}}$ be a sequence of functions in $C(\bar{\Omega})$ that are harmonic in Ω . If the u_j converge uniformly on $\partial\Omega$, then they converge uniformly on $\bar{\Omega}$ to a function $u \in C(\bar{\Omega})$ that is harmonic in Ω .

Proof. By the maximum principle, the sequence $\{u_j\}_{j \in \mathbb{N}}$ is Cauchy in the uniform topology of $C(\bar{\Omega})$. Hence, there is a continuous function $u : \bar{\Omega} \rightarrow \mathbb{R}$ to which the u_j converge uniformly on $\bar{\Omega}$. Since the u_j will also converge locally uniformly to u in Ω , the claim follows by the Harnack theorems. $\square$

Problem 2.2. Let $\Omega \subseteq \mathbb{R}^n$ be open and let $\{u_j\}_{j \in \mathbb{N}}$ be a sequence of harmonic functions in Ω such that

$$u_j \rightarrow u \quad \text{in } L^1_{\text{loc}}(\Omega).$$

Prove that there exists a harmonic function $v : \Omega \rightarrow \mathbb{R}$ equal to u almost everywhere.

Proof. Fix a compact set $K \subseteq \Omega$ and let $V \Subset \Omega$ be an open set with $V \supseteq K$. Let $\delta > 0$ be such that $B(x, \delta) \subseteq V$ for every $x \in K$. Then, for all $j, m \in \mathbb{N}$ we have

$$\left| u_j(x) - u_m(x) \right| \leq \frac{1}{\omega_n \delta^n} \int_V |u_j - u_m| \xrightarrow{j, m \rightarrow \infty} 0.$$

It follows that $\{u_j\}_{j=1}^\infty$ converges locally uniformly to a function $v : \Omega \rightarrow \mathbb{R}$ that is harmonic in Ω . Since u_j converges to u in $L^1_{\text{loc}}(\Omega)$, we are done. $\square$

Problem 2.3. Let Ω be a domain in $\mathbb{R}^n$, let $\omega \subseteq \Omega$ be open and let $K \subset \Omega$ be compact. For each $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\sup_K |u| \leq \varepsilon$$

whenever u is a harmonic function in Ω such that

- (1) $\sup_\Omega |u| \leq 1$;
- (2) $\sup_\omega |u| \leq \delta$.

Proof. We argue by contradiction. Otherwise, we can find $\varepsilon > 0$ and corresponding sequences $\{u_k\}_{k \in \mathbb{N}}$ of harmonic functions in Ω with

- (1) $\sup_\Omega |u| \leq 1$;
- (2) $\sup_\omega |u| \leq \frac{1}{k}$;

for all $k \in \mathbb{N}$ such that $|u(x_k)| > \varepsilon$ for some $x_k \in K$. Note that $\{u_k\}_{k \in \mathbb{N}}$ is equibounded in Ω . After passing to a suitable subsequence, we may assume that u_k converges locally uniformly to a harmonic function u on Ω and that $x_k \rightarrow x$ in K . Clearly, $u \equiv 0$ in ω . By the identity theorem we see that u vanishes on all of Ω . This is impossible because

$$|u(x)| = \lim_{k \rightarrow \infty} |u_k(x_k)| \geq \varepsilon > 0.$$

This completes the proof. $\square$

3. WEAK SOLUTIONS AND SOBOLEV SPACES

Unless otherwise stated, we assume for simplicity that $n \geq 3$ and $\Omega \subset \mathbb{R}^n$ is a bounded domain.

Problem 3.1. Let $u_1, u_2 \in H^1(\Omega)$ be the unique weak solutions to the problem

$$\Delta u = u \quad \text{in } \Omega$$

such that $u_1 - g_1$ and $u_2 - g_2$ belong to $H_0^1(\Omega)$, with $g_1, g_2 \in H^1(\Omega)$ given. Prove that

$$\|u_1 - u_2\|_{H^1(\Omega)} \leq C \|g_1 - g_2\|_{H^1(\Omega)}.$$

for a constant $C > 0$ independent of $u_{1,2}$ and $g_{1,2}$.

Proof. For $j = 1, 2$ we have

$$\int_\Omega \nabla u_j \cdot \nabla v + \int_\Omega u_j v = 0, \quad \forall v \in H_0^1(\Omega).$$

Especially, we have

$$\int_{\Omega} \nabla(u_1 - u_2) \cdot \nabla(u_1 - g_1 - u_2 + g_2) + \int_{\Omega} (u_1 - u_2)(u_1 - g_1 - u_2 + g_2) = 0.$$

But this implies that

$$\int_{\Omega} |\nabla(u_1 - u_2)|^2 + \int_{\Omega} (u_1 - u_2)^2 = \int_{\Omega} \nabla(u_1 - u_2) \cdot \nabla(g_1 - g_2) + \int_{\Omega} (u_1 - u_2)(g_1 - g_2).$$

Hence,

$$\begin{aligned} \|u_1 - u_2\|_{H^1(\Omega)}^2 &\leq \left(\|\nabla u_1 - \nabla u_2\|_{L^2(\Omega)} + \|u_1 - u_2\|_{L^2(\Omega)} \right) \|g_1 - g_2\|_{H^1(\Omega)} \\ &\leq 2 \|u_1 - u_2\|_{H^1(\Omega)} \|g_1 - g_2\|_{H^1(\Omega)}. \end{aligned}$$

This completes the proof. $\square$

Problem 3.2. Let $I = (a, b)$ be a bounded open interval in $\mathbb{R}$ and let $f \in L^2(I)$ have a second order weak derivative $f'' \in L^2(I)$. Prove that $f' \in L^2(I)$ exists in the weak sense.

Proof. For simplicity let us assume that $I = (0, 1)$. Let $\varphi \in C_c^\infty(I)$ be given and let $\phi \in C_c^\infty(I)$ be such that $\varphi \equiv \phi'$. Define

$$f' : I \rightarrow \mathbb{R}, \quad f'(x) := \int_0^x f''.$$

By Hölder's inequality, this function is bounded and hence belongs to $L^2(I)$. Next, we calculate

$$\begin{aligned} \int_0^1 f'(x)\varphi(x) dx &= \int_0^1 \int_0^x f''(y)\phi'(x) dy dx \\ &= \int_0^1 \int_y^1 f''(y)\phi'(x) dx dy \\ &= - \int_0^1 \phi(y)f''(y) dy \\ &= - \int_0^1 \phi'' f \\ &= - \int_0^1 \varphi' f. \end{aligned}$$

This completes the proof. $\square$

Problem 3.3. Given an open set $\Omega \subseteq \mathbb{R}^n$ and $\varepsilon > 0$, show that

$$\|\nabla u\|_{L^2(\Omega)}^2 \leq \varepsilon \|\Delta u\|_{L^2(\Omega)}^2 + \frac{1}{4\varepsilon} \|u\|_{L^2(\Omega)}^2$$

for all $u \in H_0^2(\Omega)$.

Proof. By a density argument, it is enough to prove the inequality for $u \in C_c^\infty(\Omega)$. For any such function, an integration by parts shows that

$$\begin{aligned} \|\nabla u\|_{L^2(\Omega)}^2 &\leq \int_{\Omega} |\nabla u|^2 = - \int_{\Omega} u \Delta u \leq \|\Delta u\|_{L^2(\Omega)} \|u\|_{L^2(\Omega)} \\ &= 2 \left(\sqrt{\varepsilon} \|\Delta u\|_{L^2(\Omega)} \right) \left(\frac{\|u\|_{L^2(\Omega)}}{2\sqrt{\varepsilon}} \right) \\ &\leq \varepsilon \|\Delta u\|_{L^2(\Omega)}^2 + \frac{1}{4\varepsilon} \|u\|_{L^2(\Omega)}^2 \end{aligned}$$

where we have used the inequality $2ab \leq a^2 + b^2$ in this last step. $\square$

Problem 3.4. Let Ω be a “sufficiently nice” bounded domain in $\mathbb{R}^n$. For each $\varepsilon > 0$, show that there exists a constant $C > 0$ such that

$$\|\nabla u\|_{L^2(\Omega)} \leq \varepsilon \|u\|_{H^2(\Omega)}^2 + \frac{C}{\varepsilon} \|u\|_{L^2(\Omega)}^2$$

for all $u \in H^2(\Omega)$.

Proof. Let $E : H^2(\Omega) \rightarrow H^2(\mathbb{R}^n)$ be an extension operator that acts faithfully with respect to the L^2 -norm. We may also assume that Eu always has support in an open set $V \ni \Omega$. Now, for suitable $C > 0$ we have

$$\begin{aligned} \|\nabla u\|_{L^2(\Omega)}^2 &\leq \|\nabla v\|_{L^2(\mathbb{R}^n)}^2 \leq \frac{\varepsilon}{C^2} \|\Delta v\|_{L^2(\mathbb{R}^n)}^2 + \frac{C^2}{4\varepsilon} \|v\|_{L^2(\mathbb{R}^n)}^2 \\ &\leq \frac{\varepsilon}{C^2} \|v\|_{H^2(\mathbb{R}^n)}^2 + \frac{\tilde{C}}{\varepsilon} \|u\|_{L^2(\Omega)}^2 \\ &\leq \varepsilon \|u\|_{H^2(\Omega)}^2 + \frac{\tilde{C}}{\varepsilon} \|u\|_{L^2(\Omega)}^2. \end{aligned}$$

Here, we have made use of the fact that $H^2(\mathbb{R}^n) = H_0^2(\mathbb{R}^n)$. $\square$

Problem 3.5. Let $I := (-1, 1)$ and consider the continuous function $u : (-1, 1) \rightarrow \mathbb{R}$ given by $u(x) := |x|$. Prove that $u \in W^{1,\infty}(I)$ but that u does not belong to the closure of $W^{1,\infty}(I) \cap C^\infty(I)$. Especially, this shows that $u \notin W_0^{1,\infty}(I)$.

Proof. Note that u is absolutely continuous on $(-1, 1)$ and thus has a weak derivative u' that is equal to the classical derivative of u at almost every point of $(-1, 1)$. Thus,

$$u' = \begin{cases} 1 & \text{if } x \geq 0, \\ -1, & \text{if } x \leq 0. \end{cases}$$

Since u and u' are bounded, we have $u \in W^{1,\infty}(I)$. Assume there exists a sequence $\{u_k\}_{k \in \mathbb{N}}$ in $C^\infty(I) \cap W^{1,\infty}(I)$ such that $u_k \rightarrow u$ in $W^{1,\infty}(I)$. In particular, $\{u_k\}_{k \in \mathbb{N}}$ and $\{u'_k\}_{k \in \mathbb{N}}$ are Cauchy with respect to the uniform norm on $(-1, 1)$. Hence, we can find $v, v' \in C(I)$ such that

$$u_k \xrightarrow[k \rightarrow \infty]{\text{uniformly}} v \quad \text{and} \quad u'_k \xrightarrow[k \rightarrow \infty]{\text{uniformly}} v'$$

on I . Since $u_k \rightarrow u$ and $u'_k \rightarrow u'$ in $L^\infty(I)$, we must have $u = v$ almost everywhere on I and $u' = v'$ almost everywhere on I . However, there is no continuous function $I \rightarrow \mathbb{R}$ that is equal almost everywhere to u' . $\square$

Let $a, b \geq 0$ be given and fix $1 \leq p < \infty$. The function $x \mapsto x^p$ is convex on $[0, \infty)$ and hence we see that

$$\frac{(a+b)^p}{2^p} = \left(\frac{a}{2} + \frac{b}{2}\right)^p \leq \frac{1}{2}(a^p + b^p).$$

Since $a, b \geq 0$ was arbitrary, we conclude that

$$(a+b)^p \leq 2^{p-1}(a^p + b^p), \quad \forall p \in [1, \infty), \forall a, b \geq 0. \quad (1)$$

Henceforth, we assume that $1 \leq p < \infty$ and that $I := (0, 1)$.

Problem 3.6. Show that

$$\|u\|_{L^\infty(I)} \leq 2^{1-1/p} \|u\|_{W^{1,p}(I)}$$

for all $u \in W^{1,p}(I)$.

Proof. By density, it is enough to prove the claim for $u \in C^\infty(\bar{I})$. Given $x \in I$, we have

$$\begin{aligned} |u(x)| &= \left| \int_0^1 u(x) \, dy \right| \leq \left| \int_0^1 (u(x) - u(y)) \, dy \right| + \int_\Omega |u(y)| \, dy \\ &\leq \left| \int_0^1 \int_y^x u'(t) \, dt \, dy \right| + \|u\|_{L^p(I)} \\ &\leq \int_0^1 \int_0^1 |u'(t)| \, dt \, dy + \|u\|_{L^p(I)} \\ &\leq \|u'\|_{L^p(I)} + \|u\|_{L^p(I)}. \end{aligned}$$

Applying (1) then yields the desired inequality. $\square$

Problem 3.7. Let $u \in W^{1,p}(I)$ be given. Prove that there exists a continuous function $v : \bar{I} \rightarrow \mathbb{R}$ such that $u = v$ almost everywhere on I .

Proof. Let $\{v_j\}_{j \geq 1}$ be a sequence in $C^\infty(\bar{I})$ that converges to u in $W^{1,p}(I)$ as $j \rightarrow \infty$. In particular, $\{v_j\}_{j \in \mathbb{N}}$ is Cauchy in $W^{1,p}(I)$. By the previous problem, this sequence is also Cauchy in $C(\bar{I})$. Hence, there is a continuous function $v : \bar{I} \rightarrow \mathbb{R}$ to which the v_j converge uniformly on $\bar{I}$. After passing to a subsequence, we may also assume that $v_j \rightarrow u$ pointwise almost everywhere on I . This shows that $u = v$ almost everywhere on I . $\square$

Henceforth we identify $u \in W^{1,p}(I)$ with its continuous representative $\bar{I} \rightarrow \mathbb{R}$, which we know to exist by the above. It is not hard to see that v is uniquely determined.

Problem 3.8. For any $\xi \in [0, 1]$, prove that

$$\|u\|_{L^p(I)}^p \leq 2^{p-1} \|u'\|_{L^p(I)} + 2^{p-1} |u(\xi)|^p$$

for all $u \in W^{1,p}(I)$.

Proof. Without loss of generality assume that u is continuous on $\bar{I}$. Let $\{v_j\}_{j \geq 1}$ be a sequence in $C^1(\bar{I})$ such that

- (1) $v_j \rightarrow u$ in $W^{1,p}(I)$;
- (2) $v_j \rightarrow u$ uniformly on $\bar{I}$.

Then, for every index $j \in \mathbb{N}$ and every $x \in I$:

$$\begin{aligned} |v_j(x)| &\leq \left| \int_{\xi}^x v_j' \right| + |v_j(\xi)| \leq \int_I |v_j'| + |v_j(\xi)| \\ &\leq \|v_j'\|_{L^p(I)} + |v_j(\xi)|. \end{aligned}$$

Letting $j \rightarrow \infty$, we discover that

$$|u(x)| \leq \|u'\|_{L^p(I)} + |u(\xi)|$$

for all $x \in I$. Next, we raise everything to the power p and apply (1):

$$|u(x)|^p \leq \left(\|u'\|_{L^p(I)} + |u(\xi)| \right)^p \leq 2^{p-1} \left(\|u'\|_{L^p(I)}^p + |u(\xi)|^p \right).$$

Integrating over $x \in I$ finally yields the asserted inequality. $\square$

Problem 3.9. Show that

$$\|u\|_{L^p(I)}^p \leq 2^{p-1} \left(\|u'\|_{L^p(I)}^p + \left| \int_I u \right|^p \right)$$

for all $u \in W^{1,p}(I)$.

Proof. By the previous problem, it is enough to show that there is some $\xi \in I$ such that $u(\xi) = \int_I u$. We may also assume that u is continuous on $\bar{I}$. Let x^* be the point of $\bar{I}$ at which u achieves its global maximum, and let x_* be the global minimum of u . Next, consider the function

$$v : \bar{I} \rightarrow \mathbb{R}, \quad x \mapsto u(x) - \int_I u.$$

It is not hard to check that $v(x^*) \geq 0$ and that $v(x_*) \leq 0$. The intermediate value theorem then implies the assertion. $\square$

Problem 3.10. Let $\Omega \subseteq \mathbb{R}^n$ be a bounded domain with Lipschitz-boundary and let $f \in H^1(\Omega)$. If there exists $u \in H^1(\Omega)$ such that

$$\int_{\Omega} \nabla u \cdot \nabla v = \int_{\Omega} f v, \quad \forall v \in H^1(\Omega),$$

then $\int_{\Omega} f = 0$. Conversely, if $\int_{\Omega} f = 0$, there exists $u \in H^1(\Omega)$ such that the above holds true. In fact, this u is unique up to addition by a constant.

Proof. Let $\int_{\Omega} f = 0$ and consider the space

$$S := \left\{ v \in H^1(\Omega) : \int_{\Omega} v = 0 \right\}.$$

It is not hard to check that this is a closed subspace of $H^1(\Omega)$, and hence is a Hilbert space in its own right. If $u \in S$ then we have

$$\bar{u} = \oint_{\Omega} u = 0.$$

By the Poincaré inequality, we must have $\|u\|_{H^1(\Omega)} \leq C \|\nabla u\|_{H^1(\Omega)}$ for a suitable $C > 0$ independent of $u \in S$. Hence, the bilinear form

$$\int_{\Omega} \nabla u \cdot \nabla v$$

is an inner product on S equivalent to that inherited from $H^1(\Omega)$. Thus, by the Riesz representation theorem, we can find $u \in S$ such that

$$\int_{\Omega} \nabla u \cdot \nabla v = \int_{\Omega} f v, \quad \forall v \in S.$$

It is not hard to extend this to all $v \in H^1(\Omega)$. □

4. SPECTRAL THEORY OF THE LAPLACIAN

Problem 4.1. Let $\Omega \subset \mathbb{R}^n$ be a bounded Lipschitz domain and consider Friedrich's inequality

$$\int_{\Omega} |u|^2 \leq C \int_{\Omega} |\nabla u|^2, \quad \forall u \in H_0^1(\Omega).$$

with $C > 0$ constant depending only on Ω . Characterize the best constant by an eigenvalue problem.

Proof. Let $\{u_k\}$ and $\{\lambda_k\}$ be eigenvectors and eigenvalues for the extended Laplacian A on $H^1(\Omega)$. These form an orthogonal complete family in $H_0^1(\Omega)$ and an orthonormal complete family in $L^2(\Omega)$. Furthermore, we know that $\lambda_k \nearrow \infty$ in k . By Friedrich's inequality, we know that $\lambda_1 > 0$. We know from class that for all $u \in H_0^1(\Omega)$:

$$\|\nabla u\|_{L^2(\Omega)}^2 \geq \lambda_1 \sum_{k=1}^{\infty} \left| \langle u, u_k \rangle_{L^2(\Omega)} \right|^2 = \lambda_1 \|u\|_{L^2(\Omega)}^2,$$

with equality if and only if $Au = \lambda_1 u$. Thus, the best constant is $\frac{1}{\lambda_1}$. □